# measure_and_integral

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# 1 Measurable Space

## 1.1 Measurable Space

In $\mathbb{R}$, we define the volume of an interval to be the length of it. That is, if a closed [1] interval is of the form $I = [a, b]$, then its length is $b - a$. Additionally, we want several properties such as: *additivity*, i.e. if a set is a disjoint union of intervals then the volume of that set is the sum of volumes of the intervals, *complement*, i.e. if a set has volume then the complement also has. It leads to the definitions of $\sigma$-algebra and measure.

**Definition 1 ($\sigma$-algebra)** *Let $X$ be an arbitrary set, $M$ is said to be a $\sigma$-algebra on $X$ if $M$ consists of subsets of $X$ with properties as follows:*

- $\varnothing, X \in M$
- $U \in M \implies X \setminus U \in M$ *(complement)*
- $U_1, U_2, ... \in M \implies \bigcup_{k\in\mathbb{N}} U_k \in M$ *(countable union[2])*

**Definition 2 (Measurable Space)** *Let $X$ be an arbitrary set and a $\sigma$-algebra $M$ defined on $X$. A function $\mu : M \to [0, +\infty]$ is called measure if $\mu(\varnothing) = 0$ and given a countable collection of disjoint sets $U_1, U_2, ... \in M$ then $\mu(\bigcup_{k\in\mathbb{N}} U_k) = \sum_{k\in\mathbb{N}} \mu(U_k)$. The triplet $(X, M, \mu)$ is called a measurable space. A set $U \in M$ is called a measurable set.*

## 1.2 Basis of a Measurable Space

Given an arbitrary collection of $\sigma$-algebras, the union and intersection are also $\sigma$-algebras. That enables us to talk about $\sigma$-algebra generated by a basis.

**Definition 3 ($\sigma$-algebra generated by basis)** *Let $X$ be an arbitrary set and $B$ be a collection of subsets of $X$. Define $\sigma(B)$ to be the smallest $\sigma$-algebra containing $B$, that is the intersection of all $\sigma$-algebra containing $B$. $\sigma(B)$ is also said to be the $\sigma$-algebra generated by $B$.*

**Definition 4 (Basis of a $\sigma$-algebra)** *Let $X$ be an arbitrary set and $M$ be a $\sigma$-algebra on $X$. A subset $B$ of $M$ is said to be a basis of $M$ if $M = \sigma(B)$*

As discussed earlier, to define volume of a set in $\mathbb{R}$, we will construct a $\sigma$-algebra consisting all open intervals on $\mathbb{R}$ as follows

**Definition 5 (Borel $\sigma$-algebra or Borel algebra)** *Let $(X, \mathcal{T})$ be a topological space, the Borel $\sigma$-algebra on $X$, denoted by $\mathscr{B}(X)$ is the smallest $\sigma$-algebra defined on $X$ containing all open sets.*

The Borel Measurable Space is then defined as

**Definition 6 (Borel Measurable Space on $\mathbb{R}^n$)** *$(\mathbb{R}^n, \mathscr{B}(\mathbb{R}^n), \mu)$ is the Borel Measurable Space on $\mathbb{R}^n$ if $\mu$ is defined by the volume of $n$-intervals.*

$$\mu\left(\prod_{k=1}^{n}[a_k, b_k]\right) = \prod_{k=1}^{n}(b_k - a_k)$$

[1] similarly for other types of interval
[2] or $\sigma$-union

### 1.3 Subspace of a Measurable Space

**Definition 7 (Subspace of a Measurable Space)** *Let $(X, M, \mu)$ be a measurable space and $E \in M$. Define the subspace $(E, M_E, \mu_E)$ by the restriction of $M$ and $\mu$ as follows*

- $M_E = \{U \in M : U \subseteq E\}$
- $\mu_E(U) = \mu(U)$ *if* $U \in M_E$

### 1.4 Complete Measurable Space

I haven't understood enough to give a reason for complete measurable space

**Definition 8 (Outer Measure)** *Let $(X, M, \mu)$ be a measurable space. Define outer measure $\mu_e : \mathcal{P}(X) \to [0, +\infty]$ by the infimum of the measure of all measurable covers*

$$\mu_e(E) = \inf \{\mu(S) : E \subseteq S \in M\}$$

Outer Measure satisfies several properties

- $A \subseteq B \implies |A|_e \leq |B|_e$ (subset)
- $\left|\bigcup_{k\in\mathbb{N}} E_k\right| \leq \sum_{k\in\mathbb{N}} |E_k|$ ($\sigma$-subadditivity)

**Definition 9 (Complete Measurable Space)**

**Definition 10 (Completion of a Measurable Space)**

**Definition 11 (Lebesgue Measurable Space)** [3] *Lebesgue Measurable Space on $\mathbb{R}^n$ is defined as the completion of the Borel Measurable Space on $\mathbb{R}^n$*

### 1.5 Product Measure

**Definition 12 (Product Measurable Space)** *Let $(X, M, \mu)$ and $(Y, N, \nu)$ be measurable spaces. Define the product measurable space as $(X \times Y, \sigma(M \times N), \mu \times \nu)$ where the product measure $\mu \times \nu$ is defined as*

$$(\mu \times \nu)(E_X \times E_Y) = \mu(E_X)\nu(E_Y)$$

*where $E_X \in M$ and $E_Y \in N$*

### 1.6 Image Measure

## 2 Measurable Function

**Definition 13 (Measurable Function)** *Let $(X, M, \mu)$ and $(Y, N, \nu)$ be measurable spaces. A function $f : X \to Y$ is said to be measurable if*

$$E_Y \in N \implies f^{-1}(E_Y) \in M$$

In this text, we are particularly interested in the case where $Y = [0, +\infty]$ as it is the foundation to define Lebesgue integral. Let's first define a topology on the totally ordered set $[0, +\infty]$

**Definition 14 (Order Topology)** *Given a totally ordered set $(X, <)$, the order topology $\mathcal{T}$ on $X$ is defined by the basis consists of rays as follows*

$$B = \{\{x :\in X : x < a\} : a \in X\} \cup \{\{x :\in X : b < x\} : b \in X\}$$

The Borel $\sigma$-algebra on $[0, +\infty]$ is now defined as follows

**Definition 15 (Measurable Function on Half Extended Real Line)** *Let $[0, +\infty]$ be equipped with the order topology. Define the Borel measurable space on $[0, +\infty]$ by $([0, +\infty], \mathscr{B}([0, +\infty]), \nu)$ where $\nu$ is defined by the length of intervals. Let $(X, M, \mu)$ be a measurable space. A function $f : X \to [0, +\infty]$ is said to be measurable if it is a measurable function to the Borel measurable space on $[0, +\infty]$.*

From now on, we will call *Measurable Function on Half Extended Real Line* by *Measurable Function* if there is no confusion. It is trivial to verify the proposition below.

**Proposition 1** *Let $(X, M, \mu)$ be a measurable space. A function $f : X \to [0, +\infty]$ is measurable if and only if*

$$f^{-1}(a, +\infty] \in M$$

*for all $a \in [0, +\infty]$*

[3] can also defined as the unique non-trivial measurable space on $\mathbb{R}^n$ that translation invariant

# 3 Integration

## 3.1 Simple Function

**Definition 16 (Indicator function)** *Let $E \subseteq X$, the indicator function of $E$ denoted by $\chi_E : X \to [0, +\infty]$ is defined by*

$$\chi_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{otherwise} \end{cases}$$

**Definition 17 (Non-negative simple function)** *A function is said to be simple if the set image is finite set. A non-negative simple function $\phi : X \to [0, +\infty]$ can be written as*

$$\phi = \sum_{n=1}^{N} a_n \chi_{E_n}$$

*where $N \in \mathbb{N}$, $a_n \in [0, +\infty]$, $\{E_n\}$ are pairwise disjoint, and $X = \bigcup_{n=1}^{N} E_n$*

**Definition 18 (Integral of a non-negative simple function)** *If $\phi = \sum_{n=1}^{N} a_n \chi_{E_n}$ is a non-negative simple function where $E_n$ are measurable w.r.t some measurable space $(X, M, \mu)$. We define the integral of $\phi$ as follows*

$$\int_X \phi = \sum_{n=1}^{M} a_n \mu(E_n)$$

**Theorem 1 (Monotone sequence of simple functions)** *Given any $f : \mathbb{R}^n \to [0, +\infty]$, there exists a monotone increasing sequence of simple functions $\phi_k : \mathbb{R}^n \to [0, +\infty]$ that converges pointwise to $f$*

## 3.2 Integral of a non-negative function

**Definition 19 (Integral of a non-negative function)** *Let $(X, M, \mu)$ be a measurable space and $f : X \to [0, +\infty]$ be measurable. The integral of $f$ is defined as*

$$\int_X f d\mu = \sup \left\{ \int_X g d\mu : g \leq f \right\}$$

**Definition 20 (Lebesgue integral of a non-negative function)** *Let $(\mathbb{R}^n, M, \mu)$ be the Lebesgue measurable space on $\mathbb{R}^n$ and $f : \mathbb{R}^n \to [0, +\infty]$ be measurable. The Lebesgue integral of $f$ is defined as*

$$\int_{\mathbb{R}^n} f d\mu = \sup \left\{ \int_{\mathbb{R}^n} \phi d\mu : \phi \leq f, \phi \text{ is simple} \right\}$$

**Proposition 2 (Area under the graph)** *Let $(\mathbb{R}^n, M, \mu)$ be the Lebesgue measurable space on $\mathbb{R}^n$ and $f : \mathbb{R}^n \to [0, +\infty]$ be measurable. Define the set under $f$ as follows*

$$R(f) = \{(x, y) \in \mathbb{R}^n \times \mathbb{R} : 0 \leq y < f(x)\}$$

*Then,*

$$\mu(R(f)) = \int_{\mathbb{R}^n} f d\mu$$

**Definition 21 (Integral of arbitrary function)** *Let $(\mathbb{R}^n, M, \mu)$ be the Lebesgue measurable space on $\mathbb{R}^n$ and $f : \mathbb{R}^n \to [-\infty, +\infty]$ be measurable. The Lebesgue integral of $f$ is defined as*

$$\int_{\mathbb{R}^n} f d\mu = \int_{\mathbb{R}^n} f^+ d\mu - \int_{\mathbb{R}^n} f^- d\mu$$

*if either $\int_{\mathbb{R}^n} f^+ d\mu$ and $\int_{\mathbb{R}^n} f^- d\mu$ is finite.*

**Proposition 3 (Linearity of Lebesgue integral)**

## 3.3 Convergence Theorems of a non-negative function

**Proposition 4 (Monotone Convergence Theorem)**

**Proposition 5 (Fatou Lemma)**

**Theorem 2 (Lebesgue Dominated Convergence Theorem)**

## 3.4 Product Measure Theorems

**Proposition 6 (Sectional Property for Lebesgue Measurable Space)** *Let $X = \mathbb{R}^m, Y = \mathbb{R}^n$, $E$ is a Lebesgue measurable set on $X \times Y$. For almost every $x \in X$, the set $E_x = \{y \in Y : (x, y) \in E\}$, namely section, is Lebesgue measurable on $Y$*

**Theorem 3 (Fubini Theorem)** *Let $X = \mathbb{R}^m, Y = \mathbb{R}^n$. If $f : X \times Y \to [-\infty, +\infty]$ is Lebesgue integrable, i.e. $f \in L(X \times Y)$, then*

- *For almost every $x \in X$, function $f_x : Y \to [-\infty, +\infty]$ defined by $f_x(y) = f(x, y)$ is Lebesgue integrable, i.e $f_x \in L(Y)$*
- *Function $g(x) : X \to (-\infty, +\infty)$ defined by $g(x) = \int_Y f_x(y)dy$ is Lebesgue integrable, i.e $g \in L(X)$*

*And*

$$\int_{X \times Y} f = \int_X \left[ \int_Y f_x(y)dy \right] dx$$

**Theorem 4 (Tonelli Theorem)** *Let $X = \mathbb{R}^m, Y = \mathbb{R}^n$. If $f : X \times Y \to [0, +\infty]$ is Lebesgue measurable, i.e. $f \in L(X \times Y)$, then*

- *For almost every $x \in X$, function $f_x : Y \to [-\infty, +\infty]$ defined by $f_x(y) = f(x, y)$ is Lebesgue measurable, i.e $f_x \in L(Y)$*
- *Function $g(x) : X \to (-\infty, +\infty)$ defined by $g(x) = \int_Y f_x(y)dy$ is Lebesgue measurable, i.e $g \in L(X)$*

*And*

$$\int_{X \times Y} f = \int_X \left[ \int_Y f_x(y)dy \right] dx$$

## 3.5 Change of Variables

## 3.6 Lebesgue Differentiation Theorem

**Lemma 1 (Zorn Lemma)** *Suppose a partially ordered set $P$ has the property that every chain in $P$ has an upper bound in $P$. Then the set $P$ contains at least one maximal element.*

**Lemma 2 (Axiom of Choice)** *For any set $X$ of nonempty sets, there exists a choice function $f$ that is defined on $X$ and maps each set of $X$ to an element of that set.*

**Lemma 3 (Vitali Covering Lemma)** *In a metric space, let $F$ be a collection of balls such that $\sup_{B \in F} r(B) = R < +\infty$. Then it has a (countable separable metric space) subcollection $M$ if pairwise disjoint balls such that*

$$\bigcup_{B \in F} B \subseteq \bigcup_{B_k \in M} 5B_k$$

*where $nB_k$ denotes the open ball constructed from $B_k$ by simple enlarging it $n$ times* [4].

Let $F_n$ be a collection of balls. Define a partial order on the set of all disjoint collection of balls in $F_n$ as follows: let $A, B$ be any set of disjoint balls in $F_n$, define $A \leq B$ as $A \subseteq B$.

Given any chain $A_1 \leq A_2 \leq ...$ where $A_i \in F_n$ for all $i \in N$, there is an upperbound $A = \bigcup_{i \in \mathbb{N}} A_i$. By Zorn Lemma, $F_n$ has a maximal subcollection $M$ of disjoint balls.

Let $F_1$ be defined as follows

$$F_0 = \left\{ B \in F : \frac{R}{2} < r(B) \leq R \right\}$$

Using the procedure above, construct a maximal subcollection $M_1$ of disjoint of disjoint balls in $F_1$.

Let $F_n$ be defined as follows

$$F_n = \left\{ B \in F : \frac{R}{2^{n+1}} < r(B) \leq \frac{R}{2^n} \text{ and } B \notin M_1, M_2, ..., M_{n-1} \right\}$$

Given any $B_n \in F$ such that $\frac{R}{2^{n+1}} < r(B_n) \leq \frac{R}{2^n}$, we claim that $B_n$ must intersect one of $M_1, M_2, ..., M_n$. Suppose $B$ does not intersect $M_1, M_2, ..., M_{n-1}$, then by the construction of $F_n$, $B_n \in F_n$. Furthermore, by maximality of $M_n$, $B$ either is or intersects one of ball in $M_n$.

Given any $B_n \in F$ such that $\frac{R}{2^{n+1}} < r(B_n) \leq \frac{R}{2^n}$, let $B_k \in M_k$ $(k \leq n)$ intersects $B_n$, then $\frac{R}{2^{k+1}} < r(B_k) \leq \frac{R}{2^k}$. Let $c_k, c_n$ be the centers of $B_k$ and $B_n$, let $a \in B_k \cap B_n$. For any $b \in B_n$,

[4] $\{x \in \mathbb{R}^n : ||x - x_0|| < r\} \mapsto \{x \in \mathbb{R}^n : ||x - x_0|| < nr\}$

$$
\begin{aligned}
d(c_k, b) &\leq d(c_k, a) + d(a, c_n) + d(c_n, b) \\
&\leq r(B_k) + r(B_n) + r(B_n) && (a \in B_k, a \in B_n, b \in B_n) \\
&\leq r(B_k) + \frac{4R}{2^{n+1}} && \left(r(B_n) \leq \frac{R}{2^n}\right) \\
&\leq 5r(B_k) && \left(\frac{R}{2^{n+1}} \leq \frac{R}{2^{k+1}} \leq r(B_k)\right)
\end{aligned}
$$

Hence, any point in any ball in $F$ is in one of the ball $5B_k$ where $B_k \in M_k$, $k \in \mathbb{N}$. Let $M = \bigcup_{k\in\mathbb{N}} M_k$. Given any two balls $B_m, B_n$ with $m < n$, by the construction of $F_n$, $B_n \in M_n \subseteq F_n$ then $B_n$ does not intersect $B_m$. Hence, $M$ is a pairwise disjoint subcollection of $F$ such that given any $b \in B \in F$, then $b \in \bigcup_{B_k \in M} 5B_k$

If the metric space is separable, let $M_Q = \{Q \cap B_k : B_k \in M\}$ where $Q$ is a dense set. Elements of $M_Q$ is non-empty and pairwise disjoint, hence is there a bijection from $M$ to $M_Q$ Furthermore, by Axiom of Choice, there is a bijection from $M_Q$ into a countable set $Q$. Hence, $M$ is a countable set.

In case of Lebesgue measure $\mu$ for $\mathbb{R}^n$, we can write $M = \bigcup_{k\in\mathbb{N}} B_k$, then

$$\mu(M) = \sum_{k\in\mathbb{N}} \mu(B_k) = \lim_{N\to+\infty} \sum_{k=1}^{N} \mu(B_k)$$

Given any $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$\sum_{k=1}^{N} \mu(B_k) > \mu(M) - \epsilon \geq \frac{1}{5}\mu\left(\bigcup_{B\in F} B\right) - \epsilon$$

That is, a bounded Lebesgue measurable set $A$ in $\mathbb{R}^n$ is covered by a collection of balls $F$ (not necessary bounded in radius), given any $\epsilon > 0$, there is a finite subcollection $M$ of pairwise disjoint balls of $F$ such that

$$\sum_{B_k \in M} \mu(B_k) \geq \frac{1}{5}\mu(A) - \epsilon$$

**Lemma 4 (Hardy-Littlewood Maximal Function)** *Given a Lebesgue measurable function $f : \mathbb{R}^n \to \mathbb{R}$, Hardy-Littlewood maximal function defined on ball is as follows*

$$f^*(x) = \sup_{r\,>\,0} \frac{1}{\mu(\mathcal{B}_r(x))} \int_{\mathcal{B}_r(x)} |f(y)|dy$$

*Some properties of Hardy-Littlewood maximal function:*

1. *$f^*$ is lower semicontinuous, i.e. $E_t = \{x \in \mathbb{R}^n : f^*(x) > t\}$ is open for all $\alpha \in \mathbb{R}$*
2. *$\mu(E_t) \leq \frac{c}{t}\int_{\mathbb{R}^n} |f(y)|dy$ where $c \in \mathbb{R}$*

(Part 1) Given any $\alpha > 0$, let $x_1 \in E_t = \{x \in \mathbb{R}^n : f^*(x) > t\}$ that is, there exists $r > 0$ such that

$$\frac{1}{\mu(\mathcal{B}_r(x_1))} \int_{\mathcal{B}_r(x_1)} |f(y)|dy > \alpha t$$

where $\alpha > 1$. We will prove that there exists an open ball $\mathcal{B}_\epsilon(x_1)$ such that for every $x_2 \in \mathcal{B}_\epsilon(x_1)$, $f^*(x_2) > t$, that implies the set $E_t$ is open.

Let $\epsilon > 0$, for any $x_2 \in \mathcal{B}_\epsilon(x_1)$, then $\mathcal{B}_r(x_1) \subseteq \mathcal{B}_{r+\epsilon}(x_2)$. Therefore

$$
\begin{aligned}
t &< \frac{1}{\alpha}\frac{1}{\mu(\mathcal{B}_r(x_1))} \int_{\mathcal{B}_r(x_1)} |f(y)|dy \\
&\leq \frac{1}{\alpha}\frac{\mu(\mathcal{B}_{r+\epsilon}(x_2))}{\mu(\mathcal{B}_r(x_1))}\frac{1}{\mu(\mathcal{B}_{r+\epsilon}(x_2))} \int_{\mathcal{B}_{r+\epsilon}(x_2)} |f(y)|dy
\end{aligned}
$$

Choose $\epsilon$ small enough such that $\frac{\mu(\mathcal{B}_{r+\epsilon}(x_2))}{\mu(\mathcal{B}_r(x_1))} < \alpha$, then

$$\frac{1}{\mu(\mathcal{B}_{r+\epsilon}(x_2))}\int_{\mathcal{B}_{r+\epsilon}(x_2)}|f(y)|dy > t$$

That is, $f^*(x_2) > t$. Hence, there exists $\epsilon > 0$ such that for all $x_2 \in \mathcal{B}_\epsilon(x_1)$, $f^*(x_2) > t$. $E_t$ is open for any $t$.

(Part 2) For each $x \in E_t$, let $B_x$ be a ball centered at $x$ such that $\frac{1}{\mu(B_x)}\int_{B_x}|f(y)|dy > t$. The collection $\{B_x : x \in E_t\}$ covered $E_t$, by Vitali Covering Lemma, given any $\epsilon > 0$, there is a finite subcollection $M_\epsilon$ of pairwise disjoint $B_k$ such that

$$\sum_{B_k \in M_\epsilon}\mu(B_k) \geq \frac{1}{5}\mu(E_t) - \epsilon$$

Therefore,

$$\begin{aligned}
\mu(E_t) &\leq 5\sum_{B_k \in M_\epsilon}\mu(B_k) + 5\epsilon \\
&< 5\sum_{B_k \in M_\epsilon}\left(\frac{1}{t}\int_{B_k}|f(y)|dy\right) + 5\epsilon && \text{(by construction of } B_k\text{)} \\
&\leq \frac{5}{t}\int_{\mathbb{R}^n}|f(y)|dy + 5\epsilon && (B_k \text{ pairwise disjoint})
\end{aligned}$$

The statement is true for all $\epsilon > 0$, hence

$$\mu(E_t) \leq \frac{5}{t}\int_{\mathbb{R}^n}|f(y)|dy$$

**Lemma 5 (Density of Continuous Function)** *Given $f \in L(\mathbb{R}^n)$ and any $\epsilon > 0$, there exists a continous function $g$ with compact support such that*

$$\int_{\mathbb{R}^n}|f(x) - g(x)|dx < \epsilon$$

**Theorem 5 (Lebesgue Differentiation Theorem)** *Given $f \in L(\mathbb{R}^n)$, then for almost every $x \in \mathbb{R}^n$*

$$\limsup_{B \searrow x}\frac{1}{\mu(B)}\left|\int_B [f(y) - f(x)]dy\right| = 0$$

*where $B \searrow x$ denotes $B = \mathcal{B}_r(x)$ and $r \to 0^+$*

Let $D_t$ be defined as

$$D_t = \left\{x \in \mathbb{R}^n : \limsup_{B \searrow x}\frac{1}{\mu(B)}\left|\int_B [f(y) - f(x)]dy\right| > t\right\}$$

We will prove that $\mu(D_t) = 0$ for all $t > 0$, that will conclude the Lebesgue Differentiation Theorem.

Let $g$ be a continous function, and any ball $B$ centers at $x$, then

$$\begin{aligned}
\left|\int_B [f(y) - f(x)]dy\right| &\leq \int_B |f(y) - f(x)|dy \\
&\leq \int_B |f(y) - g(y)|dy + \int_B |g(y) - g(x)|dy + \mu(B)|g(x) - f(x)|
\end{aligned}$$

Hence,

$$\begin{aligned}
\limsup_{B \searrow x}\frac{1}{\mu(B)}\left|\int_B [f(y) - f(x)]dy\right| &\leq \limsup_{B \searrow x}\frac{1}{\mu(B)}\left(\int_B |f(y) - g(y)|dy + \int_B |g(y) - g(x)|dy + \mu(B)|g(x) - f(x)|\right) \\
&\leq \limsup_{B \searrow x}\frac{1}{\mu(B)}\int_B |f(y) - g(y)|dy + \limsup_{B \searrow x}\frac{1}{\mu(B)}\int_B |g(y) - g(x)|dy + \limsup_{B \searrow x}|g(x) - f(x)|
\end{aligned}$$

As $g$ is continuous, $\limsup_{B \searrow x}\frac{1}{\mu(B)}\int_B |g(y) - g(x)|dy = 0$, then

$$\limsup_{B \searrow x}\frac{1}{\mu(B)}\left|\int_B [f(y) - f(x)]dy\right| \leq \limsup_{B \searrow x}\frac{1}{\mu(B)}\int_B |f(y) - g(y)|dy + |g(x) - f(x)|$$

Therefore,

$$\begin{aligned}
D_t &\subseteq \left\{x \in \mathbb{R}^n : \limsup_{B \searrow x} \frac{1}{\mu(B)} \int_B |f(y) - g(y)| dy > \frac{t}{2}\right\} \cup \left\{x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2}\right\} \\
&\subseteq \left\{x \in \mathbb{R}^n : \sup_{B} \frac{1}{\mu(B)} \int_B |f(y) - g(y)| dy > \frac{t}{2}\right\} \cup \left\{x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2}\right\} \\
&= \left\{x \in \mathbb{R}^n : (f-g)^*(x) > \frac{t}{2}\right\} \cup \left\{x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2}\right\}
\end{aligned}$$

As the two sets are measurable

$$\mu(D_t) \leq \mu\left(\left\{x \in \mathbb{R}^n : (f-g)^*(x) > \frac{t}{2}\right\}\right) + \mu\left(\left\{x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2}\right\}\right)$$

By property of Hardy-Littlewood maximal function,

$$\mu\left(\left\{x \in \mathbb{R}^n : (f-g)^*(x) > \frac{t}{2}\right\}\right) \leq \frac{2c}{t} \int_{\mathbb{R}^n} |f(y) - g(y)| dy$$

By Chebyshev inequality,

$$\mu\left(\left\{x \in \mathbb{R}^n : |g(x) - f(x)| > \frac{t}{2}\right\}\right) \leq \frac{2}{t} \int_{\mathbb{R}^n} |f(y) - g(y)| dy$$

Therefore,

$$\mu(D_t) \leq \frac{2(c+1)}{t} \int_{\mathbb{R}^n} |f(y) - g(y)| dy$$

Given any $t > 0$ and $\epsilon > 0$, we can choose $g$ close enough to $f$ such that $\int_{\mathbb{R}^n} |f(y) - g(y)| dy < \frac{t\epsilon}{2(c+1)}$, then $\mu(D_t) < \epsilon$. Hence $\mu(D_t) = 0$ for all $t > 0$. Then

$$D = \left\{x \in \mathbb{R}^n : \limsup_{B \searrow x} \frac{1}{\mu(B)} \left|\int_B [f(y) - f(x)] dy\right| > 0\right\} = \bigcup_{k \in \mathbb{N}} D_{1/k}$$

Then $\mu(D) = 0$, that is $\limsup_{B \searrow x} \frac{1}{\mu(B)} \left|\int_B [f(y) - f(x)] dy\right| = 0$ for almost every $x \in \mathbb{R}^n$